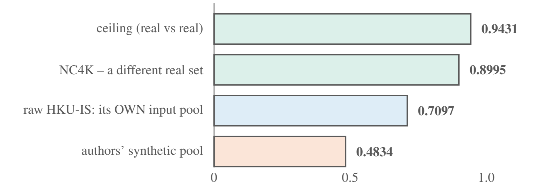
EXP A3 — synthetic vs real

[scope: generator] [scope: method-independent]

How far do LAKE-RED's renders sit from the real target distribution – and does the metric that was used to say so survive an honest control?

(a) MANIFOLD COVERAGE — k-NN recall vs the real target set, dinoL518, $k=5$

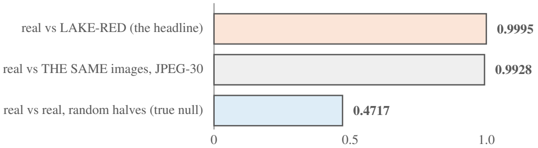


synthetic covers 0.4834 where its *own input pool* covers 0.7097

a paired within-content loss of -31.9%; the render moves the pool *away* from the target

the comparison that matters is against LAKE-RED's own input, not against the ceiling — it is the only one that holds content fixed

(b) BUT THE METRIC THAT USED TO CARRY THIS IS NEAR-VACUOUS — clipL224



re-encoding the *identical* images separates them from themselves at 0.9928

within **0.0067** of the headline, on content that is identical by construction

and the estimator manufactures effects: on a *true null* the in-sample d is **+0.6867** while held-out it is **+0.0318**

OUTCOME

The finding survives; the metric that used to carry it does not. The coverage loss is real and passes 3/3 embedders. The AUC framing is withdrawn — a near-1.0 separation is not evidence that two image sets differ distributionally.

a distributional characterisation, not a training-utility bound

"far, therefore useless" is a non-sequitur